Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/12/2004
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Project Title:	Extinction and Instability Mechanisms of Polymerization Fronts		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Polymers		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	01/06/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Meiburg, Eckart (University of California, Santa Barbara)		
Grant/Contract No.:	NAG-1701		
Performance Goal No.:			
Performance Goal Text:			

Recent advances in polymer chemistry have led to the development of monomers and initiation agents providing propagating polymerization fronts driven by the exothermicity of the polymerization reaction and the transport of heat from the polymerized product to the monomer. The purpose of this study is to determine the mechanisms of extinction and instability and thereby determine means to obtain more useful product material at earth gravity and µg. Experiments are being conducted in two distinct geometries, specifically Hele-Shaw cells and round tubes. Our prior experiments in gas combustion in round tubes of varying diameter have demonstrated two distinct extinction limits due to these processes; it will be determined if the same applies to polymer fronts. In addition to µg experiments, in the Hele-Shaw cell varying the angle of the cell relative to the horizontal will be used to assess buoyancy effects _ though with limitations because a minimum cell thickness and thus minimum Rayleigh number is required to avoid heat-loss induced quenching. The experiments will measure propagation rates and front shapes using direct video, laser-induced fluorescence (LIF) imaging, a technique not previously used in studies of polymerization fronts, along with temperature

Task Description:	measurements using thermocouples and laser interferometry, and flow properties via particle tracking and laser Doppler velocimetry. The polymerized products will be characterized using NMR to determine the conversion efficiency and Gel Permeation Chromatography to measure the product molecular weight distribution. Also, prior experiments in aqueous autocatalytic chemical reaction fronts in Hele-Shaw cells have shown a previously unknown manifestation of the Saffman-Taylor instability due to surface tension effects, even though the reactants and products are completely miscible in each other. We are determining if such surface tension effects exist in polymer fronts as well. A companion numerical modeling effort is being performed. A novel aspect of these studies is the inclusion of Korteweg stresses that enable modeling of surface tension-like forces in regions of steep concentration or temperature gradients at the interface between miscible fluids. Another important aspect of the modeling study is the inclusion of heat losses to walls, which are anecdotally know to cause extinction of polymer fronts but have not been the subject of systematic evaluation in prior literature. The use of polymerization processes based on this mode of polymerization has many applications including rapid curing of polymers without external heating, uniform curing of thick samples, solventless preparation of some polymers, and filling/sealing of structures having cavities of arbitrary shape without having to heat the structure externally. One important limitation of this process is that the fronts extinguish when they try to propagate through channels that are too narrow (probably due to conductive heat losses) or too wide (for unknown reasons, which we propose to be convective heat losses driven by buoyancy-induced flow.) Even when extinction does not occur, convective and buoyant instabilities can affect the structure and properties of the resulting polymerized materials as well as the propagation rates of the fronts. Microgravity experiments are an indispensable aspect of this project because polymer fronts are very likely subject to quenching by buoyancy-driven mechanisms, and buoyancy also leads to a wide variety of instabilities, which may overwhelm effects due to surface tension, viscosity difference, and thermal expansion/contraction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have successfully produced polymerization fronts in water-based solutions at atmospheric pressure without boiling using dilute solutions of highly reactive acrylic acid (AA) monomer and ammonium persulfate (AP) initiator. Prior studies of frontal polymerization required a high-temperature solvent such as DMSO, or very high pressures to avoid boiling, despite the fact that water at ambient pressure is the most convenient medium and the most important for practical applications (because of the cost and environmental issues associated with other solvents). High temperatures were previously needed for polymerization fronts because fronts with peak temperatures less than 100°C had not been identified _ at such low temperatures the fronts would extinguish. By developing a better understanding of extinction and instability mechanisms in polymer fronts, we have established low-temperature water-based polymerization reactions supporting propagating fronts. The propagation speeds and quenching limits of the polymer fronts have been determined for a range of tube diameters and AP and AA concentrations for tubes that are uninsulated, in open air, (subject to heat losses due to buoyant convection), in a stirred water bath (to obtain near-isothermal conditions), and insulated (to obtain near-adiabatic conditions). In uninsulated tubes the extinction limit due to conductive heat loss to tube walls was observed at a Peclet number (Pe, defined as the Sd/D, where S is the front propagation speed, d the tube diameter and D the thermal diffusivity of the mixture) of about 5 - comparable to predictions from combustion theory. The tube diameter was found to affect extinction limits (e.g. the concentration of AP at limit), but the value of Pe at the limit was found to be almost constant. At the limit, the front speed increased as the tube diameter decreased. Also, the thermal boundary conditions were found to affect propagation rates and extinction limits. Experiments conducted in a stirred water bath did not produce smoothly propagating fronts. For the range of conditions in which there is a smooth front, the heat loss to the water bath was greater than the heat generation of the front, thus extinguishing. Experiments conducted with insulated test tubes (to obtain near-adiabatic conditions) demonstrated steadily propagating fronts with a significantly lower front speed and concentration of AA at the limit. In insulated tubes, limiting values of Pe were much lower, in fact no well-defined limits were observed. Instead very slowly propagating (less than 10 μm/s) but still steady fronts were observed. The Pe of the slowest measurable front in the insulated tubes was about 1.3. Temperature profiles for front propagation in insulated and uninsulated tubes were obtained. A dramatic difference in the heat loss profiles of the two systems was observed, as expected, indicating that our insulation was effective and that heat losses were the cause of extinction limits in the uninsulated tubes. In the computational effort, in order to investigate the influence of viscosity differences onto the density-driven instability of two miscible fluids in a vertical Hele-Shaw cell, we conducted a linear stability analysis. Dispersion relations are obtained for different Rayleigh numbers, viscosity ratios, and interfacial thickness parameters of the base concentration profile. The analysis employs the three-dimensional Stokes equations, and the results are compared with those obtained from the variable density and viscosity Hele-Shaw equations. While the growth rate does not depend on which of the two fluids is the more viscous one, the maxima of the eigenfunctions are always seen to shift towards the less viscous fluid. For every parameter combination, the dominant instability mode is found to be three-dimensional. With increasing viscosity ratio, the instability is uniformly damped. For a fixed viscosity ratio, both the growth rate and the most unstable wavenumber increase monotonically with the Rayleigh number, until they asymptotically reach a plateau. Surprising findings are obtained regarding the effects of varying the interface thickness. At higher viscosity ratios the largest growth rates and unstable wavenumbers are observed for intermediate thicknesses. This demonstrates that for variable viscosities thicker interfaces can be more unstable than their thinner counterparts, in contrast to the constant viscosity case. The reason behind this behavior can be traced to the influence of the gap width on the vertical extent of the perturbation eigenfunctions. For thick interfaces, the eigenfunction can reside almost entirely within the interfacial region. In that way, the perturbation maximum is free to shift towards the less viscous fluid, i.e., into a locally more unstable environment. In contrast, for thin interfaces the eigenfunction is forced to extend far into the viscous fluid, which leads to an overall stabilization. While the Hele-Shaw analysis also captures this `optimal' growth for intermediate interface thicknesses, the growth rates differ substantially from those obtained from the full Stokes equations. Compared to the Hele-Shaw results, growth rates obtained from the modified Brinkman equation are seen to yield better quantitative agreement with the Stokes results.
Bibliography Type:	Description: (Last Updated: 01/29/2007)
Abstracts for Journals and Proceedings	Demuth, R.; Meiburg, E. "Chemical Fronts in Hele-Shaw Cells: Linear Stability Analysis Based on the Three-Dimensional Stokes Equations" Phys. Fluids 15, 59 , Jan-2003